

Operating Systems

Top 100 Interview Questions & Answers

A Comprehensive Guide for Technical Interviews

Topics Covered:

Introduction & Basics • Process Management • Threads & Concurrency
CPU Scheduling • Process Synchronization • Deadlocks
Memory Management • Virtual Memory • File Systems • I/O & Security

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1. Introduction and Basics

Q1. What is an Operating System (OS)?

Answer: An Operating System is system software that acts as an intermediary between a user of a computer and the computer hardware. It manages computer hardware resources and provides common services for computer programs.

Q2. What are the primary purposes of an OS?

Answer: The two primary purposes are: 1) Resource Management (managing CPU time, memory space, file storage, I/O devices) and 2) Abstraction/Convenience (providing a user-friendly interface to hardware).

Q3. What is the Kernel?

Answer: The kernel is the core component of an OS that manages system resources. It is the program that is always running in memory and has complete control over everything in the system.

Q4. Differentiate between Monolithic Kernel and Microkernel.

Answer: In a monolithic kernel, all OS services run along with the main kernel thread, making it faster but larger and prone to crashes if one service fails. In a microkernel, user services and kernel services are kept in separate address spaces, making it smaller and more stable, but slower due to message passing.

Q5. What is Bootstrapping?

Answer: Bootstrapping (or booting) is the process of starting or restarting a computer. A small piece of code called the bootstrap program (stored in ROM) initializes the system and loads the OS kernel into memory.

Q6. What is a System Call?

Answer: A system call is a programming interface provided by the OS to user applications. It allows user-level processes to request privileged services from the OS kernel, such as hardware access or process creation.

Q7. What is Multiprogramming?

Answer: Multiprogramming is a technique to keep multiple programs in main memory at the same time and keep the CPU busy by switching between them when one program is waiting for I/O operations.

Q8. What is Multitasking (Time-Sharing)?

Answer: Multitasking is a logical extension of multiprogramming. The CPU executes multiple jobs by switching among them very frequently, giving users the illusion that they are interacting with multiple programs at once.

Q9. What is Multiprocessing?

Answer: Multiprocessing refers to a computer system's ability to support more than one processor (CPU) to execute multiple processes simultaneously, improving throughput and

reliability.

Q10. What is Spooling?

Answer: Spooling (Simultaneous Peripheral Operations On-Line) is a process in which data is temporarily held to be used and executed by a device, program, or the system. For example, sending documents to a printer queue.

2. Process Management

Q11. What is a Process?

Answer: A process is a program in execution. It is an active entity that requires resources like CPU time, memory, files, and I/O devices to accomplish its task.

Q12. What are the different states of a process?

Answer: New (being created), Ready (waiting to be assigned to a processor), Running (instructions are being executed), Waiting/Blocked (waiting for an event like I/O), and Terminated (execution finished).

Q13. What is a Process Control Block (PCB)?

Answer: A PCB is a data structure maintained by the OS for every process. It stores process state, program counter, CPU registers, scheduling information, memory-management information, and I/O status.

Q14. What is a Context Switch?

Answer: A context switch is the process of saving the state of the currently running process and loading the state of the next process to be executed. It introduces overhead as the system does no useful work during the switch.

Q15. Differentiate between a Program and a Process.

Answer: A program is a passive entity (a file on a disk containing instructions), whereas a process is an active entity (a program loaded into memory and executing). A program can spawn multiple processes.

Q16. What does the `fork()` system call do?

Answer: `fork()` creates a new process by duplicating the calling process. The new process is called the child process, and the original is the parent. It returns 0 to the child and the child's PID to the parent.

Q17. What is an Orphan Process?

Answer: An orphan process is a child process whose parent has terminated or finished execution without waiting for the child to terminate. In UNIX-like systems, orphans are adopted by the `init` process.

Q18. What is a Zombie Process?

Answer: A zombie process is a process that has completed execution but still has an entry

in the process table. This occurs when the parent process has not yet read the child's exit status using the `wait()` system call.

Q19. What is Interprocess Communication (IPC)?

Answer: IPC is a mechanism that allows processes to communicate and synchronize their actions. It is essential for cooperating processes.

Q20. What are the two primary models of IPC?

Answer: Shared Memory (processes read/write to a shared region of memory) and Message Passing (processes communicate by sending and receiving messages over a communication channel).

3. Threads and Concurrency

Q21. What is a Thread?

Answer: A thread is the basic unit of CPU utilization, consisting of a thread ID, a program counter, a register set, and a stack. It is a lightweight process that shares the code, data, and OS resources with other threads within the same process.

Q22. What is the difference between a Process and a Thread?

Answer: A process is heavyweight, has its own memory space, and takes longer for context switching. A thread is lightweight, shares memory space with sibling threads, and has faster context switching.

Q23. Differentiate User-level threads and Kernel-level threads.

Answer: User-level threads are managed entirely by user-level libraries and the kernel is unaware of them (fast to create/switch, but a blocking system call blocks the whole process). Kernel-level threads are managed directly by the OS (slower, but one blocked thread doesn't block others).

Q24. What are the Multithreading Models?

Answer: Many-to-One (many user threads to one kernel thread), One-to-One (each user thread maps to one kernel thread), and Many-to-Many (many user threads multiplexed to a smaller or equal number of kernel threads).

Q25. What is a Thread Control Block (TCB)?

Answer: Similar to a PCB, a TCB is a data structure used to store information about a single thread, including thread state, program counter, register values, and a pointer to the PCB of the process it belongs to.

Q26. What is Thread Cancellation?

Answer: Thread cancellation is the task of terminating a thread before it has completed. It can be asynchronous (one thread immediately terminates the target thread) or deferred (the target thread periodically checks if it should terminate).

4. CPU Scheduling

Q27. What is CPU Scheduling?

Answer: CPU scheduling is the process of determining which process in the ready queue gets to use the CPU next, aiming to maximize CPU utilization and throughput while minimizing turnaround and waiting times.

Q28. What is the role of the Dispatcher?

Answer: The dispatcher is the module that gives control of the CPU to the process selected by the short-term scheduler. It handles context switching, switching to user mode, and jumping to the proper location in the program.

Q29. Differentiate Preemptive and Non-preemptive scheduling.

Answer: Non-preemptive scheduling allows a running process to keep the CPU until it voluntarily releases it (terminates or waits). Preemptive scheduling allows the OS to forcefully interrupt a running process to allocate the CPU to another process.

Q30. What is Throughput?

Answer: Throughput is the number of processes that are completed per time unit. It is a metric used to evaluate scheduling algorithms.

Q31. What is Turnaround Time?

Answer: Turnaround time is the total time taken to execute a particular process. It is the interval from the time of submission to the time of completion (waiting time + execution time).

Q32. What is Waiting Time?

Answer: Waiting time is the sum of the periods spent waiting in the ready queue. The scheduling algorithm directly impacts this metric.

Q33. What is Response Time?

Answer: Response time is the time from the submission of a request until the first response is produced. It is crucial in interactive systems.

Q34. Explain FCFS (First-Come, First-Served) Scheduling.

Answer: The process that requests the CPU first is allocated the CPU first. It is non-preemptive, simple, but can suffer from the Convoy Effect (short processes waiting for a long process to finish).

Q35. Explain SJF (Shortest Job First) Scheduling.

Answer: The CPU is assigned to the process that has the smallest next CPU burst. It can be preemptive or non-preemptive. It gives the minimum average waiting time but requires predicting future burst times.

Q36. What is SRTF (Shortest Remaining Time First)?

Answer: SRTF is the preemptive version of SJF. If a new process arrives with a CPU burst length less than the remaining time of the currently executing process, the CPU is pre-empted.

Q37. Explain Round Robin (RR) Scheduling.

Answer: Designed for time-sharing systems, each process gets a small unit of CPU time (time quantum). After this time has elapsed, the process is preempted and added to the end of the ready queue.

Q38. Explain Priority Scheduling.

Answer: A priority is associated with each process, and the CPU is allocated to the process with the highest priority. Equal-priority processes are scheduled in FCFS order. It can lead to starvation.

Q39. What is Multilevel Queue Scheduling?

Answer: The ready queue is partitioned into several separate queues (e.g., foreground and background). Processes are permanently assigned to one queue, and each queue has its own scheduling algorithm. Queues are also scheduled against each other.

5. Process Synchronization

Q40. What is Process Synchronization?

Answer: It is the coordination of execution of multiple processes in a multi-process environment to ensure that they access shared data and resources in a controlled predictable manner, preventing data inconsistency.

Q41. What is a Critical Section?

Answer: A critical section is a segment of code in a process where shared resources (like common variables, files) are accessed. Only one process should execute in its critical section at a time.

Q42. What are the requirements for a solution to the Critical Section problem?

Answer: 1) Mutual Exclusion (only one process in critical section), 2) Progress (if no process is in critical section, a process wanting to enter cannot be postponed indefinitely), and 3) Bounded Waiting (limit on how many times other processes can enter before a waiting process gets its turn).

Q43. What is Mutual Exclusion (Mutex)?

Answer: Mutual exclusion is a property of concurrency control ensuring that if a process is executing in its critical section, no other processes can be executing in their critical sections simultaneously.

Q44. What is a Semaphore?

Answer: A semaphore is an integer variable accessed through two standard atomic operations: `wait()` (decrement) and `signal()` (increment), used to solve various synchronization problems.

Q45. What is the difference between a Mutex and a Semaphore?

Answer: A mutex is a locking mechanism used to serialize access to a section of code (ownership based). A semaphore is a signaling mechanism (can be released by a different thread) used to manage concurrent access to a pool of resources.

Q46. What is a Binary Semaphore?

Answer: A binary semaphore is a semaphore with an integer value that can range only between 0 and 1. It behaves similarly to a mutex lock.

Q47. What is a Monitor?

Answer: A monitor is a high-level synchronization construct that abstracts away the complexity of semaphores. It encapsulates shared data structures, procedures, and synchronization between concurrent procedure invocations within a single object.

Q48. What is the Bounded-Buffer (Producer-Consumer) Problem?

Answer: A classic synchronization problem where a Producer generates data and places it in a buffer, and a Consumer takes data out. They must be synchronized so the producer doesn't add to a full buffer and the consumer doesn't take from an empty one.

Q49. What is the Dining Philosophers Problem?

Answer: A problem illustrating synchronization issues involving N philosophers sitting at a table with N chopsticks. A philosopher needs two chopsticks to eat. It highlights the challenges of avoiding deadlock and starvation when allocating multiple resources.

6. Deadlocks

Q50. What is a Deadlock?

Answer: A deadlock is a situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process in the set.

Q51. What are the four necessary conditions for Deadlock?

Answer: 1) Mutual Exclusion, 2) Hold and Wait, 3) No Preemption, and 4) Circular Wait. All four must hold simultaneously for a deadlock to occur.

Q52. What is the Resource-Allocation Graph?

Answer: It is a directed graph used to precisely describe deadlocks. Vertices represent processes and resources. Edges represent requests and assignments. A cycle in this graph may indicate a deadlock.

Q53. What are the methods for handling Deadlocks?

Answer: 1) Deadlock Prevention (ensure one of the 4 conditions cannot hold), 2) Deadlock Avoidance (OS checks state before allocating), 3) Deadlock Detection and Recovery, and 4) Ignore the problem (Ostrich Algorithm, common in standard OS like Windows/Linux).

Q54. What is Deadlock Prevention?

Answer: A set of methods to ensure that at least one of the necessary conditions (Mutual exclusion, hold and wait, no preemption, circular wait) cannot hold, thus preventing deadlocks structurally.

Q55. What is Deadlock Avoidance?

Answer: Requires that the OS be given additional information about which resources a process will request. The OS uses this info to dynamically decide if a resource allocation will leave the system in a "safe state."

Q56. What is a Safe State?

Answer: A state is safe if the system can allocate resources to each process in some order (a safe sequence) such that every process can execute to completion without causing a deadlock.

Q57. What is Banker's Algorithm?

Answer: A resource allocation and deadlock avoidance algorithm that tests for safety by simulating the allocation of predetermined maximum possible amounts of all resources, and then makes an "s-state" check to test for possible deadlock conditions before deciding whether allocation should be allowed to continue.

Q58. What is Deadlock Recovery?

Answer: If a deadlock is detected, the system can recover by either aborting one or more processes to break the circular wait, or preempting resources from some processes and giving them to others.

7. Memory Management

Q59. What is Main Memory?

Answer: Main memory (RAM) is a volatile storage that stores data and programs currently being executed by the CPU. The CPU can directly access it.

Q60. Differentiate Logical Address and Physical Address.

Answer: A logical address (virtual address) is generated by the CPU during execution. A physical address is an actual location in the main memory. The set of all logical addresses is the logical address space.

Q61. What is the Memory Management Unit (MMU)?

Answer: The MMU is a hardware device that maps logical addresses to physical addresses at run time.

Q62. What is Swapping?

Answer: Swapping is a mechanism in which a process can be swapped temporarily out of main memory (or move) to secondary storage (disk), and make that memory available to other processes. It is later swapped back to resume execution.

Q63. What is Contiguous Memory Allocation?

Answer: Each process is contained in a single contiguous section of memory. It suffers from external fragmentation.

Q64. What is External Fragmentation?

Answer: Total memory space is enough to satisfy a request or to reside a process in it, but it is not contiguous, so it cannot be used.

Q65. What is Internal Fragmentation?

Answer: Memory allocated to a process is slightly larger than the requested memory. The difference between allocated and requested memory is wasted within the partition.

Q66. What is Paging?

Answer: Paging is a memory management scheme that eliminates the need for contiguous allocation of physical memory. Physical memory is divided into fixed-size blocks called frames, and logical memory into blocks of the same size called pages.

Q67. What is a Translation Look-aside Buffer (TLB)?

Answer: A TLB is a fast associative hardware cache used to store recently used page-table entries to speed up the translation of logical to physical addresses, reducing the memory access time.

Q68. What is Segmentation?

Answer: Segmentation is a memory management technique in which memory is divided into variable-sized blocks called segments, representing logical units of a program (like functions, objects, data arrays), unlike fixed-size pages.

8. Virtual Memory

Q69. What is Virtual Memory?

Answer: Virtual memory is a technique that allows the execution of processes that are not completely in memory. It abstracts main memory into a very large, uniform array of storage, separating logical memory from physical memory.

Q70. What is Demand Paging?

Answer: A system where a page is not loaded into main memory until it is actually needed (demanded) during program execution, saving memory space and reducing I/O.

Q71. What is a Page Fault?

Answer: A page fault occurs when a program attempts to access a page that is part of its virtual memory space but is not currently loaded in the physical main memory.

Q72. What is the Page Replacement process?

Answer: When a page fault occurs and there are no free frames in memory, the OS must

swap out an existing page to make room for the new one. It uses page replacement algorithms to choose which page to evict.

Q73. What is FIFO Page Replacement?

Answer: First-In-First-Out. The OS replaces the oldest page in memory. It is simple but can suffer from Belady's Anomaly.

Q74. What is Belady's Anomaly?

Answer: A phenomenon in which increasing the number of page frames allocated to a process actually increases the number of page faults. It occurs in FIFO page replacement.

Q75. What is LRU Page Replacement?

Answer: Least Recently Used. The algorithm replaces the page that has not been used for the longest period of time. It is a very good algorithm and does not suffer from Belady's anomaly.

Q76. What is Thrashing?

Answer: Thrashing occurs when a computer spends more time paging (swapping pages in and out of memory) than executing instructions. It severely degrades system performance.

9. File Systems

Q77. What is a File?

Answer: A file is a named collection of related information that is recorded on secondary storage. It is the smallest allotment of logical secondary storage.

Q78. What are File Attributes?

Answer: Attributes provide information about a file, such as Name, Identifier, Type, Location, Size, Protection (read/write/execute flags), Time, Date, and User identification.

Q79. What is Sequential Access?

Answer: A file access method where data is read or written in order, one record after another, from beginning to end (like magnetic tape).

Q80. What is Direct (Random) Access?

Answer: A file access method where files are viewed as a numbered sequence of blocks or records. The program can jump to any record directly without reading preceding records (like disk drives).

Q81. What is a Directory?

Answer: A directory is a container or logical grouping of files. It provides a way to structure and organize files on a disk.

Q82. Explain Contiguous File Allocation.

Answer: Each file occupies a set of contiguous blocks on the disk. It provides excellent read performance but suffers from external fragmentation and file growth difficulties.

Q83. Explain Linked File Allocation.

Answer: Each file is a linked list of disk blocks, which may be scattered anywhere on the disk. It solves fragmentation but random access is very slow.

Q84. Explain Indexed File Allocation.

Answer: Brings all pointers together into one location called an index block. Each file has its own index block containing addresses of its disk blocks, supporting direct access without external fragmentation.

Q85. What is an Inode in UNIX?

Answer: An Inode (Index Node) is a data structure in a Unix file system that describes a filesystem object such as a file or a directory. It stores file attributes and disk block locations.

10. I/O and Disk Management

Q86. What is the difference between Polling and Interrupts?

Answer: In polling, the CPU constantly checks the status of an I/O device to see if it needs attention (wastes CPU cycles). In interrupts, the device sends a hardware signal to the CPU when it needs attention, freeing the CPU to do other work.

Q87. What is Direct Memory Access (DMA)?

Answer: DMA is a feature that allows certain hardware subsystems to access main system memory independently of the CPU. The CPU sets up the transfer, and the DMA controller takes over, interrupting the CPU only when done.

Q88. What is Buffering?

Answer: A buffer is a memory area that stores data being transferred between two devices or between a device and an application, absorbing mismatches in data transfer speeds.

Q89. What is Spooling vs Buffering?

Answer: Buffering handles overlapping I/O of one job. Spooling manages overlapping I/O of many jobs (e.g., spooling a printer queue so programs don't have to wait for the printer).

Q90. What is Disk Scheduling?

Answer: Disk scheduling is done by the OS to schedule I/O requests arriving for the disk to minimize seek time (the time for the disk arm to move to the desired cylinder).

Q91. Explain FCFS Disk Scheduling.

Answer: First-Come, First-Served. Requests are serviced in the order they arrive. Fair but doesn't provide the fastest service.

Q92. Explain SSTF Disk Scheduling.

Answer: Shortest Seek Time First. Selects the request with the minimum seek time from the current head position. May cause starvation of some requests.

Q93. Explain SCAN (Elevator) Algorithm.

Answer: The disk arm starts at one end of the disk and moves toward the other end, servicing requests as it reaches each cylinder, until it gets to the other end, then reverses direction.

Q94. Explain C-SCAN Algorithm.

Answer: Circular SCAN. Like SCAN, but when the head reaches the other end, it immediately returns to the beginning of the disk without servicing any requests on the return trip, providing a more uniform wait time.

Q95. What is RAID?

Answer: Redundant Array of Independent Disks. A technology that combines multiple physical disk drive components into one or more logical units for data redundancy, performance improvement, or both.

11. Security, Protection, and Miscellaneous

Q96. What is the difference between Security and Protection?

Answer: Protection refers to internal mechanisms that control the access of programs, processes, or users to resources defined by a computer system. Security is a broader concept that defends the system against external and internal attacks.

Q97. What is the Principle of Least Privilege?

Answer: Programs, users, and systems should be given just enough privileges to perform their tasks, and no more. This limits the damage if a compromise occurs.

Q98. What is an Access Matrix?

Answer: A general model of protection. The rows represent domains (subjects), and the columns represent objects. Each entry specifies the set of access rights that a domain has over an object.

Q99. What is a Real-Time Operating System (RTOS)?

Answer: An OS designed to process data as it comes in, typically without buffer delays. Processing must occur within specified time constraints or the system will fail.

Q100. What is a Distributed Operating System?

Answer: An OS that manages a group of independent computers and makes them appear to be a single computer to users. It handles communication and resource sharing across

a network.